

SUMMER INTERNSHIP PROJECT REPORT



Image Data Augmentation and Image Classification Using Python

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Project Guide

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Acknowledgement

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I would also like to acknowledge the providers of the Cat-Dog image dataset and the MNIST handwritten digit dataset, which served as the primary data sources for implementing image processing, dataset preparation, and clustering techniques in this project. I also acknowledge the developers of Python and its associated libraries, including Pandas, NumPy, Matplotlib, NumPy, Matplotlib, PyTorch, Torchvision, Scikit-learn, and SciPy whose tools greatly facilitated the analysis conducted in this study.

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Abstract

Image processing and image classification have become essential components of modern Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision systems. With the increasing availability of digital images, automated methods for analyzing and classifying visual information have gained significant importance in various fields such as healthcare, security, agriculture, autonomous vehicles, and industrial automation. However, raw image data often contains noise, variations in size, orientation, illumination, and other inconsistencies that can negatively affect the performance of machine learning models. Therefore, image preprocessing and data augmentation play a crucial role in improving the quality and diversity of image datasets.

This project focuses on implementing various image processing and augmentation techniques using Python libraries such as OpenCV, PyTorch, Torchvision, NumPy, Matplotlib, and Scikit-learn. The project demonstrates how images can be loaded, transformed, resized, translated, rotated, and converted into different formats to prepare them for machine learning applications. Furthermore, the project introduces dataset preparation techniques using the Cat-Dog image dataset and demonstrates how data can be efficiently loaded through the ImageFolder and DataLoader classes provided by PyTorch.

In addition to image preprocessing, the project explores unsupervised machine learning through the application of the K-Means clustering algorithm on the MNIST handwritten digit dataset. The clustering performance is evaluated using the Macro F1 Score, providing insights into the effectiveness of unsupervised learning for image analysis. Overall, the project provides practical experience in image processing, computer vision, machine learning, and data preparation techniques that are widely used in modern AI applications.

Introduction

Artificial Intelligence (AI) has transformed the way computers interact with the world. One of the most important branches of AI is Computer Vision, which enables machines to interpret and understand visual information from images and videos. Applications of computer vision include facial recognition, autonomous vehicles, medical diagnosis, object detection, surveillance systems, and industrial automation.

Image classification is a supervised learning task where images are assigned labels according to their content. The success of image classification depends heavily on the quality of data used for training machine learning models. Real-world images often contain variations in position, orientation, lighting conditions, and scale. These challenges can negatively affect model performance if appropriate preprocessing techniques are not applied.

Image augmentation is a widely used technique for generating additional training samples from existing images through transformations such as rotation, translation, flipping, and scaling. Data augmentation helps improve model generalization and reduces overfitting.

During the first two weeks of internship training, the following topics were covered:

- Python Programming (NumPy and Pandas)
- Data Visualization using Matplotlib
- Statistics for Data Science
- Machine Learning Fundamentals
- Introduction to Deep Learning
- LLM fundamentals
- Model Evaluation Metrics
- Clustering Algorithms
- K-Means Clustering

The knowledge acquired during the training period was applied to implement this project successfully.

Objectives

The primary objectives of this project are:

1. To understand fundamental image processing techniques using OpenCV.
2. To perform image augmentation for improving dataset diversity.
3. To prepare image datasets for machine learning and deep learning applications.
4. To create and manage image datasets using PyTorch utilities.
5. To implement K-Means clustering on image data and evaluate its performance.

METHODOLOGY

❖ Project Overview

The project titled "Image Data Augmentation and Image Classification Using Python" focuses on understanding and implementing fundamental computer vision and machine learning techniques for image analysis. The project aims to demonstrate how raw image data can be processed, transformed, and prepared for machine learning applications using modern Python-based tools and libraries.

The project begins with basic image processing operations such as image loading, grayscale conversion, resizing, translation, and rotation using the OpenCV library. These preprocessing techniques help improve image quality, reduce computational complexity, and prepare images for further analysis. To increase dataset diversity and improve model robustness, image augmentation techniques such as resizing and random horizontal flipping are applied through Torchvision transformation pipelines.

A Cat-Dog image dataset is used to demonstrate dataset preparation and management using PyTorch's ImageFolder and DataLoader utilities. These tools enable efficient loading, labeling, and batch processing of image data, which are essential components of machine learning and deep learning workflows.

In addition to image preprocessing and augmentation, the project explores unsupervised machine learning using the MNIST handwritten digit dataset. K-Means clustering is applied to identify patterns and group similar digit images into clusters. The effectiveness of the clustering process is evaluated using the Macro F1 Score, providing insights into the performance of unsupervised learning techniques on image data.

The project integrates concepts from Computer Vision, Machine Learning, Data Science, and Artificial Intelligence, providing practical exposure to real-world image analysis tasks. The knowledge and skills gained through this project establish a strong foundation for advanced topics such as Deep Learning, Convolutional Neural Networks (CNNs), Object Detection, and Image Recognition systems.

❖ Tools and Libraries Used

Several Python libraries were used throughout the project:

- Python Programming Language
- Jupyter Notebook
- NumPy
- OpenCV (cv2)

- Matplotlib
- PyTorch.
- Torchvision
- Scikit-learn
- SciPy
- K-Means Clustering
- F1 Score

❖ Dataset Description

The project utilizes two datasets.

Cat-Dog Dataset

The Cat-Dog dataset contains images belonging to two classes:

- Cats
- Dogs

The dataset is organized into separate folders corresponding to each class. Such organization enables automatic label generation using PyTorch's ImageFolder class.

The dataset is primarily used for:

- Image preprocessing
- Image augmentation
- Dataset preparation
- Deep learning applications

MNIST Dataset

The MNIST dataset is one of the most widely used benchmark datasets in machine learning. It contains grayscale images of handwritten digits from 0 to 9.

Characteristics:

- 70,000 images
- 10 digit classes
- Image size: 28×28 pixels

The MNIST dataset was used for clustering analysis and performance evaluation.

❖ Data Preprocessing

Image Loading

The first step involved loading images using OpenCV. The dimensions of each image were examined to understand the image structure represented by height, width, and color channels.

Understanding image dimensions is important because machine learning models require consistent image formats.

Grayscale Conversion

Color images consist of three channels: Red, Green, and Blue (RGB). In many applications, color information is not necessary.

Therefore, images were converted to grayscale using OpenCV.

- Reduced computational complexity
- Faster processing
- Simplified feature extraction

Image Saving

After processing, images were saved to disk using OpenCV's image writing function.

This demonstrated:

- Image storage techniques
- File handling operations
- Preservation of transformed images

Image Translation

Translation refers to shifting an image along the horizontal or vertical direction.

Transformation matrices were applied to move images:

- Left
- Right
- Up

- Down

Translation is commonly used in data augmentation because it helps machine learning models become less sensitive to object position.

Image Resizing

Machine learning models require images with fixed dimensions.

Images were resized to standardized dimensions using OpenCV and Torchvision transformation functions.

Advantages:

- Uniform dataset structure
- Reduced computational requirements
- Improved training efficiency

Image Rotation

Images were rotated by 90 degrees counterclockwise.

Rotation helps create additional training examples and enables models to recognize objects regardless of orientation.

Data Augmentation Pipeline

A transformation pipeline was implemented using Torchvision's Compose function.

The pipeline included:

1. Resize images to 255×255 pixels.
2. Apply random horizontal flipping.
3. Convert images into tensor format.

This automated preprocessing workflow improves dataset diversity and helps prevent overfitting during model training.

Dataset Preparation

The ImageFolder class was used to organize image data.

Features of ImageFolder:

- Automatic image loading
- Automatic label assignment
- Integration with transformation pipelines

Images belonging to different categories were stored in separate folders, allowing efficient dataset management.

DataLoader Construction

PyTorch's DataLoader was used to load image batches efficiently.

Benefits include:

- Faster data loading
- Batch processing
- Memory optimization
- Shuffling of training data

DataLoader is an essential component in deep learning workflows.

MNIST Dataset Processing

The MNIST dataset was imported and prepared for machine learning experiments.

Each image was flattened into a feature vector suitable for clustering analysis.

The resulting dataset consisted of:

- Input features
- Corresponding digit labels

K-Means Clustering

K-Means is an unsupervised learning algorithm used to partition data into clusters.

The algorithm works by:

1. Selecting cluster centers.
2. Assigning observations to nearest centers.
3. Updating cluster centers.
4. Repeating until convergence.

For the MNIST dataset:

- Number of clusters = 10
- Each cluster represented a digit class.

Performance Evaluation

After clustering, cluster labels were mapped to actual digit labels.

The Macro F1 Score was used as the evaluation metric.

The Macro F1 Score combines:

- Precision
- Recall

and provides a balanced assessment of classification performance.

Methodology Flowchart

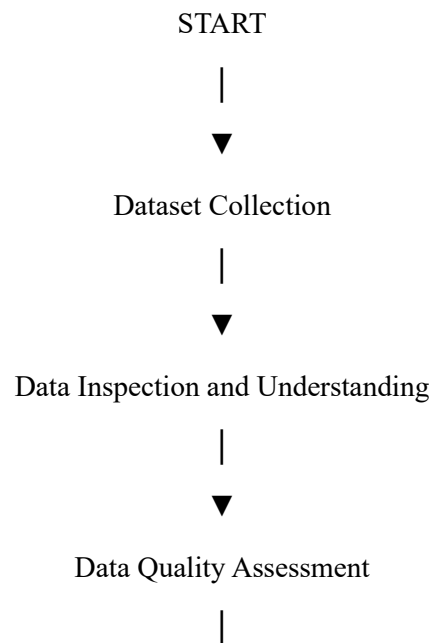




Image Preprocessing



Image Augmentation



Dataset Preparation



DataLoader Construction



Feature Extraction



K-Means Clustering



Performance Evaluation



Results Interpretation



Conclusion



END

Source Code Respository:

<https://github.com/apratim-673/Image-Data-Augmentation-and-Image-Classification-Using-Python1.git>

Data Analysis and Results

❖ Data Analysis

The analysis was conducted on two image datasets: the Cat-Dog image dataset and the MNIST handwritten digit dataset. The Cat-Dog dataset was used to demonstrate image preprocessing, augmentation, dataset preparation, and batch loading, whereas the MNIST dataset was utilized for machine learning analysis using the K-Means clustering algorithm.

Image Preprocessing Analysis

Several preprocessing techniques were applied to improve data quality and ensure consistency among images.

Grayscale Conversion

Color images represented in RGB format were converted to grayscale to reduce computational complexity.

The grayscale intensity value is computed as:

$$Gray = 0.299R + 0.587G + 0.114B$$

where:

- R = Red channel intensity
- G = Green channel intensity
- B = Blue channel intensity

This transformation reduced the dimensionality of image data while preserving important visual information.

Image Resizing

Images in the dataset had varying dimensions. Therefore, all images were resized to:

$$255 \times 255$$

pixels using torchvision transformations.

This standardization ensures uniform input dimensions for machine learning and deep learning models.

Data Augmentation

To increase dataset diversity and improve model robustness, image augmentation techniques were applied.

The augmentation pipeline included:

- Resizing
- Random Horizontal Flipping
- Tensor Conversion

Mathematically, horizontal flipping transforms a pixel coordinate (x, y) into:

$$(x', y') = (W - x - 1, y)$$

where:

- W = image width

Data augmentation helps reduce overfitting and improves the generalization capability of learning algorithms.

Machine Learning Analysis

K-Means Clustering

K-Means is an unsupervised learning algorithm used to partition observations into K clusters.

The objective function minimized by K-Means is:

$$J = \sum_{i=1}^K \sum_{x_j \in C_i} \|x_j - \mu_i\|^2$$

where:

- K = number of clusters

- C_i = cluster i
- μ_i = centroid of cluster i
- x_j = data point

For the MNIST dataset:

$$K = 10$$

since the dataset contains ten digit classes (0–9).

Cluster Assignment

Each observation was assigned to the nearest centroid using Euclidean distance:

$$d(x, \mu) = \sqrt{\sum_{i=1}^n (x_i - \mu_i)^2}$$

The clustering process was repeated iteratively until convergence.

Performance Evaluation

Since K-Means is an unsupervised learning algorithm, cluster labels were mapped to actual digit labels for evaluation.

Precision

Precision measures the proportion of correctly predicted observations among all observations assigned to a class.

$$Precision = \frac{TP}{TP + FP}$$

where:

- TP = True Positives
- FP = False Positives

Recall

Recall measures the proportion of actual observations correctly identified.

$$Recall = \frac{TP}{TP + FN}$$

where:

- FN = False Negatives

F1 Score

The F1 Score is the harmonic mean of Precision and Recall.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Macro F1 Score

For multiclass evaluation, the Macro F1 Score is calculated as:

$$Macro\ F1 = \frac{1}{N} \sum_{i=1}^N F1_i$$

where:

- N = number of classes
- $F1_i$ = F1 Score of class i

The following results were obtained during the analysis:

Operation	Result
Image Loading	Successful
Grayscale Conversion	Successful
Image Translation	Successful
Image Rotation	Successful
Image Resizing	Successful
Data Augmentation	Successful
Dataset Preparation	Successful
DataLoader Construction	Successful
K-Means Clustering	Successful
Macro F1 Score	0.79

The image preprocessing techniques successfully standardized the dataset and improved image consistency. Data augmentation increased the diversity of training samples and enhanced the robustness of the data preparation pipeline.

The K-Means clustering algorithm effectively grouped similar handwritten digit images into clusters. The obtained Macro F1 Score of:

0.79 (79%)

indicates that the clustering model achieved satisfactory performance in identifying underlying patterns within the MNIST dataset.

Comparative Analysis of Models

Since the present project implemented only the K-Means clustering algorithm, a direct comparison with other machine learning models was not performed. However, future studies may compare K-Means with algorithms such as:

Model	Type	Evaluation Metric
K-Means	Unsupervised	Macro F1 Score
Logistic Regression	Supervised	Accuracy, Precision, Recall
Random Forest	Supervised	Accuracy, F1 Score
Support Vector Machine (SVM)	Supervised	Accuracy, Precision
Convolutional Neural Network (CNN)	Deep Learning	Accuracy, Loss

Among these models, CNNs generally achieve the highest accuracy for image classification tasks because they automatically learn hierarchical image features.

Conclusion

This project provided comprehensive exposure to image processing, data augmentation, dataset preparation, and machine learning techniques. Various preprocessing methods such as grayscale conversion, resizing, translation, and rotation were applied to improve image quality and consistency. Data augmentation techniques increased dataset diversity and prepared images for deep learning applications.

The project also demonstrated the use of PyTorch's ImageFolder and DataLoader classes for efficient dataset management. Furthermore, the implementation of K-Means clustering on the MNIST dataset highlighted the potential of unsupervised learning in image analysis. The achieved Macro F1 Score of approximately 0.79 indicates satisfactory clustering performance.

Overall, the project successfully illustrates the importance of preprocessing, augmentation, and evaluation in modern Computer Vision, Machine Learning, and Artificial Intelligence systems. The knowledge and skills gained through this project provide a strong foundation for future research and development in image classification and deep learning.

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